

# Polar Equations and Graphs



## Recognizing polar graphs

1 Identify the graph of each polar equation:

**a**  $r = 3$

**b**  $\theta = \frac{\pi}{6}$

**c**  $r = 2 + 2\cos \theta$

**d**  $r = 4\sin 3\theta$

2 For the limaçon  $r = 1 + 2\cos \theta$ :

**a** Explain how you know it has an inner loop.

**b** Find the values of  $\theta$  in  $[0, 2\pi)$  at which the curve passes through the pole.

3 For the rose  $r = 5\sin 2\theta$ :

**a** State the number of petals.

**b** State the maximum value of  $|r|$  and the smallest positive  $\theta$  at which it occurs.

4 Determine whether the graph of  $r = 2 + \sin \theta$  passes through the pole, and state the type of limaçon.